Allocation & In-Stock Tracker

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Synthetic data.

What it is

Nine stores are holding the same style. Some are days from running out, others will never sell what they have. **This works out which is which, and what should move.** One intimate apparel style across a full bra size range, nine stores, twelve weeks, which comes to 2,160 rows of store × size × week. Every figure on the four analysis tabs is a live formula pointed back at that one data tab, nothing pasted in as a value. Change a threshold and everything downstream re-runs off the new value.

The four measures, and what each one is for

Measure	How it is calculated	The question it answers
Sell-Through %	Cumulative units sold ÷ cumulative units received, to date	Was the original allocation the right size? Low means too much was sent.
Weeks of Supply	On hand ÷ trailing 4-week average weekly sales	How long does what's in the store last? Puts a small store and a large one on one scale.
In-Stock %	Share of weeks so far that ended with stock on hand	Could the customer actually buy it? Catches the store that sold out early and looks great.
Size-curve variance	Half the sum of the gaps between inventory mix and sales mix	Is the stock in the right sizes? Reads as the share of units sitting in the wrong ones.

What the data shows at week 12 (floor 2 weeks, ceiling 12 weeks, curve broken above 20%)

Store	Weeks of supply	Sell-through	In-stock	Size-curve variance	Reading
S01 Toronto - Downtown	1.1	92%	98%	33%	Below the weeks-of-supply floor, size curve broken
S02 Toronto - North	4.4	73%	100%	12%	Healthy
S03 Mississauga	7.4	63%	100%	31%	Size curve broken
S04 Ottawa	4.5	73%	100%	15%	Healthy
S05 Montreal	3.5	78%	100%	19%	Healthy
S06 Quebec City	23.4	34%	100%	9%	Above the ceiling
S07 Calgary	5.7	68%	100%	10%	Healthy
S08 Vancouver	2.2	85%	99%	19%	Healthy
S09 Halifax	12.3	57%	90%	56%	Above the ceiling, size curve broken

The call the workbook makes

17 transfers, 281 units. The buy on this style closed after week 8, so nothing more is coming in and moving stock between stores is the only lever left. Each line takes the smaller of what the sending store can spare above the 12-week ceiling and what the receiving store needs to get to the 6-week target. Below 6 units it gets suppressed entirely, because the freight costs more than the move is worth. The largest six:

Size	From	Its weeks of supply	To	Its weeks of supply	Units
34B	Quebec City	23.1	Toronto - Downtown	0.6	**39**
34C	Quebec City	24.5	Toronto - Downtown	0.1	**37**
36B	Quebec City	22.0	Toronto - Downtown	1.0	**29**
36C	Quebec City	28.9	Vancouver	1.5	**25**
32B	Quebec City	26.0	Toronto - Downtown	0.8	**20**
36D	Quebec City	21.3	Toronto - Downtown	0.3	**19**

Three things the size-curve tab catches that a total-units view does not

A store can be in stock and still be broken. Halifax holds 12.3 weeks of cover, barely over the ceiling, and reads 90% in stock. But the variance index is 56%. It was allocated an even spread across all twenty sizes against demand that concentrates in 34B to 36C, so it's holding units it can't sell and missing the ones it can.

Over-stocked isn't the same problem as wrong-curve. Quebec City is the worst store in the fleet on quantity, 23.4 weeks of supply and 34% sold through, but the variance index is only 9%. It got sent the right shape and far too much of it. That's a buy-quantity error, and you fix it by moving units out, not by re-cutting the curve.

A starved store hides its own demand. Toronto Downtown is at 1.1 weeks and 92% sold through. Because it kept running out, the trailing sales rate understates what it could have sold, which then shrinks the transfer the sheet recommends it. That's why in-stock % sits beside weeks of supply rather than underneath it.

Every threshold above lives in a single named cell on the transfer tab: the floor, the ceiling, the rebalance target, the smallest quantity worth shipping and the review week. Change one and the whole workbook re-runs. Tab 5 sets out every formula in plain English.